Spurious Predictability Under Anomaly Conditioning: How a Structural Embedding Leak Survives Walk-Forward Validation

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Abstract

We document spurious predictability in delay-embedding-based financial forecasting and trace it to its root cause. Testing four anomaly detectors across seven asset classes, we found forecast accuracy substantially higher on days flagged as locally atypical, a result that survived walk-forward cross-validation, permutation testing, and an out-of-sample holdout. The effect was an artifact of a one-step index error: the embedding's most recent component was numerically identical to the first step of the future return being predicted. We show analytically why this leak concentrates on atypical-flagged days, turning a small index error into a large, regime-specific illusion of predictability, and confirm the diagnosis algebraically, with a null control on synthetic noise, an independent re-implementation, and a power test showing the corrected pipeline still detects genuine signal. All four detectors were affected identically. We propose a null-noise-plus-surrogate-data diagnostic that would have caught this error in hours, and recommend it as standard practice.

Keywords: spurious predictability, data leakage, delay embedding, analog forecasting, surrogate data, anomaly detection

1. Introduction

Short-horizon financial return forecasting via historical-analog methods, delay-embedding recent returns into a low-dimensional state space, locating the k nearest historical analogs, and averaging their realized continuations,

draws on a long-standing technique with roots in the Analog Ensemble literature from atmospheric and space-weather forecasting [7] and in the broader nonlinear-dynamics tradition of state-space reconstruction [25, 16]. A natural extension of this idea, pursued across a large and active recent literature, is to condition forecast confidence or forecast usage on an auxiliary novelty or anomaly-detection score computed on the same embedded state. The intuition is simple: states the model has rarely seen before, measured by some notion of atypicality in the embedded space, might be more or less predictable than familiar ones.

This paper began as an attempt to formalize and test that intuition properly. Using a diffusion-model reconstruction score, and, for robustness, three classical novelty detectors (Isolation Forest, Local Outlier Factor, and Mahalanobis distance in PCA space) as the atypicality signal, we found across seven asset classes, an equity index, two energy benchmarks, two precious metals, and two cryptocurrencies, that forecast accuracy conditional on high atypicality was substantially and significantly higher than conditional on low atypicality. The result held up under nested walk-forward evaluation, formal permutation testing [10], and a genuine single-shot 20% holdout that had never been touched during model development. It even motivated a dynamical-systems explanation involving local Lyapunov exponents, which itself replicated on a second holdout.

The result was wrong. Section 4 describes how we found out and, more importantly, why none of the safeguards above caught it. We think the diagnostic protocol that eventually exposed the error, and that we recommend other researchers run before reporting a similar finding, is this paper's most useful contribution.

This kind of failure is not unique to our pipeline or to financial applications of machine learning specifically. Harvey et al. [12] documented the scale of factor proliferation in the asset-pricing literature and the role data mining plays in it, and Hou et al. [14] showed that a large share of published anomalies fail to replicate under more conservative testing. What distinguishes the case reported here is that the spurious effect did not survive sloppy validation; it survived careful validation, including a form of out-of-sample testing that is often treated as close to definitive. That is the part we think is worth documenting in detail.

1.1. Contributions

Off-by-one and look-ahead errors are a long-recognized hazard in time-series pipelines on their own; our contribution is not the existence of such an error but the specific mechanism by which it interacts with anomaly conditioning. We show, both empirically and analytically, that a one-step index error in a delay-embedding pipeline, one that by itself would produce only a small, roughly uniform accuracy inflation, is mechanically amplified by exactly the kind of unsupervised novelty or anomaly-detection conditioning this literature uses to isolate "interesting" forecasting regimes, because the detector's atypicality score and the leaked component of the target are driven by the same underlying quantity. The result is a large, regime-concentrated illusion of predictability that survives methods normally considered strong safeguards against overfitting: walk-forward cross-validation, permutation testing, and a genuine out-of-sample holdout. We verify this diagnosis through six checks of increasing pipeline complexity, ranging from a closed-form algebraic proof to a null-hypothesis control on synthetic data with a provably known absence of structure. To be precise about what this buys us: these are six checks at different levels of abstraction, not six independent sources of evidence, since checks 1 and 3 below are two representations of the same underlying fact. We show that the error affected all four structurally different detectors identically, which establishes that the apparent finding was a property of the shared embedding convention rather than of any one detection method. Finally, we propose a lightweight, two-part diagnostic protocol, a pure-noise null control together with AAFT surrogate-data testing, that would have caught this error immediately, and that we recommend as a standard pre-submission check for this class of method.

2. Related Work

Delay embedding and analog forecasting. The analog-forecasting technique underlying this investigation, locating the k nearest historical states to the current embedded state and pooling their verifying continuations, is structurally identical to the Analog Ensemble (AnEn) framework developed in the atmospheric-forecasting literature. Hamill and Whitaker [11] introduced the underlying analog-correction technique for precipitation reforecasts; Delle Monache et al. [7] extended it with a Kalman-filter/analog post-processing scheme, and Delle Monache et al. [8] formalized the probabilistic AnEn method as a way of generating an ensemble forecast directly

from a single deterministic trajectory. Alessandrini et al. [1] demonstrate AnEn's practical value for short-term wind-power forecasting, illustrating the framework's reach beyond its original precipitation application. The state-space reconstruction underlying the embedding itself follows the standard delay-coordinate approach of Takens [25] and the broader nonlinear time-series methodology surveyed in Kantz and Schreiber [16]. Nearest-neighbor and analog methods have a long history in financial forecasting on their own terms; what the broader project this paper is drawn from attempted, and what we believe is less common, is to carry over the AnEn framework specifically, under that name and with its characteristic combination of embedding, causal analog selection, and ensemble pooling, from its native meteorological setting to financial return series.

Spurious predictability and leakage in financial machine learning. The integration of machine learning into empirical finance has expanded the set of models and features available to researchers, but standard validation methods were largely developed for approximately i.i.d. data and can fail under the temporal dependence and causal-ordering constraints that financial time series impose [18]. Kapoor and Narayanan [17] document how similar leakage failures have undermined reproducibility across machine-learning-based science more broadly, outside finance specifically, which suggests the failure mode we document here is a special case of a more general problem rather than one confined to financial applications. The closest precedent to this paper's diagnostic contribution is Nikolopoulos [21], who proposes a falsification framework for financial ML pipelines that tests complete modeling workflows against synthetic reference distributions with known predictability, a goal closely related to the null-noise control used here, though developed independently and applied to a different class of method. On the statistical side, the multiplicity concerns raised by testing many model and detector variants on the same panel are formalized by White [29]'s Reality Check for data snooping and by the garden of forking paths framework of Gelman and Loken [9], applied directly to financial-anomaly research by Coqueret [6]. The community-wide review of forecasting practice by Petropoulos et al. [22] likewise stresses that rigorous, bias-free out-of-sample evaluation is the field's central methodological obligation, independent of model choice; the failure mode we document is one concrete way that obligation can be violated while every individual evaluation step still appears to have been followed correctly. Bergmeir et al. [4] show, conversely, that standard (non-blocked) cross-validation can in fact be valid for purely autoregressive prediction tasks

under conditions weaker than commonly assumed; the failure mode here is not in tension with that result, since it does not arise from using an invalid validation scheme on a sound pipeline, but from a feature-construction defect that contaminates every fold, training and held-out alike, identically, which no choice of validation scheme, valid or otherwise, is positioned to catch.

Surrogate-data testing and the chaos-in-markets debate. The second component of our diagnostic protocol draws directly on the surrogate-data methodology introduced by Theiler et al. [27] for testing nonlinearity against a matched-linear null, later refined by Schreiber and Schmitz [23, 24] with the Iterative Amplitude-Adjusted Fourier Transform (IAAFT), which matches the target spectrum more closely than the AAFT surrogates used here at higher computational cost; we chose AAFT because several hundred surrogates had to be generated per test, and its approximate spectral match was sufficient for the question asked (Sections 4.2 and 4.5). This methodology underlies a decades-long, still-unsettled literature on whether financial return series exhibit genuine deterministic chaos or linear stochastic structure that merely resembles it, critically surveyed in Barnett and Serletis [3]. Studies applying surrogate-data and related nonlinearity tests to individual equity indices have reached conflicting conclusions depending on denoising and test choice: Inglada-Pérez [15] finds no clear evidence of chaos across four major indices, while Vogl and Roetzel [28] report an S&P 500 return series consistent with roughly equal parts stochastic and deterministic chaos once the data is wavelet-denoised first. That contrast underscores that this remains an open methodological question rather than a settled one. The lesson we take from that literature, that apparent nonlinear structure in a detector statistic has to be checked against a matched linear surrogate null before it can be attributed to genuine dynamics, is precisely the lesson our own investigation initially failed to apply to its own core predictability claim, and not only to its exploratory dynamical-systems explanation.

3. Data and Baseline Method

3.1. Data

Daily close prices for seven assets, downloaded from Yahoo Finance: Brent crude (`BZ=F`), the S&P 500 index (`^GSPC`), gold (`GC=F`), silver (`SI=F`), WTI crude (`CL=F`), Bitcoin (`BTC-USD`), and Ethereum (`ETH-USD`), using the maximum available history per asset (8–20 years depending on listing date).

3.2. Delay embedding and analog selection

Log returns $r_t = \log(P_t/P_{t-1})$ are delay-embedded into $\mathbb{R}^M$ with $M = 3$, $\tau = 1$, each component divided by a single scale factor, the standard deviation of r_t over the full series, $\sigma = \mathrm{std}(r)$: $x_t = (r_t, r_{t-\tau}, r_{t-2\tau})/\sigma$. Section 5 examines the consequence of this full-sample (rather than causal, growing-window) scale choice directly. We index origins by t throughout, and switch to the index j only in Section 4.4, where j denotes an origin's position in the pipeline's internal array rather than its calendar time, to match the pipeline's own variable names exactly where the algebra depends on them. For a forecast origin, the $K = 7$ nearest historical analogs are selected causally, by Euclidean distance in the diffusion model's hidden-layer feature space described in Section 3.3 (not raw embedding distance); this analog selection rule is held fixed across all four atypicality detectors described below, only the atypicality score itself varies by detector. Analog search excludes a Theiler window of 10 days around the origin to avoid trivial temporal auto-matches [26]. The forecast is the sign of the mean $H = 3$-day forward return realized by the selected analogs.

3.3. Atypicality detectors

Four detectors score each origin's atypicality, and we refer to them throughout by shorthand names that reflect what each one responds to, without claiming they respond identically or through the same functional form. The first, which we label "diffusion-ELBO" for brevity, is the training objective of a small denoising diffusion model [13]: a 2-layer MLP with hidden width 32, trained with Adam (learning rate 2×10^{-3}, batch size 128, weight decay 10^{-5}, 50 epochs) causally on the embedded history up to the current origin and retrained every 20 folds on an expanding window. Concretely, for a query point we sample noise at all 50 diffusion steps, compute the network's noise-prediction squared error at each step, $\|\hat{\epsilon}_\theta(x_t, t) - \epsilon\|^2$ with $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon$, and average over all 50 steps; this is the L_{simple} training objective of Ho et al. [13], a reweighted surrogate for the true variational bound rather than the ELBO itself, which we flag explicitly here since the two are related but not identical. We chose this score over a plain autoencoder's reconstruction error because it aggregates reconstruction difficulty over the full noise schedule rather than a single point estimate at one noise level; it is, in this sense, a more complete density-sensitive novelty score, at the cost of more training compute than an autoencoder would need. We do not claim the diffusion architecture is necessary for the qualitative result, and Table 1 and

Table 3 show it is not: the same pattern, before correction, and its disappearance, after correction, are common to three classical detectors that do not involve a diffusion model at all. The remaining three detectors are Isolation Forest [20] with 100 trees, Local Outlier Factor [5] with 30 neighbors, and Mahalanobis distance in a $\min(M, 3)$-component PCA-reduced space with Ledoit-Wolf shrinkage [19]; at the baseline $M = 3$, this is a full-rank rotation rather than a dimensionality reduction (exact Mahalanobis distance is invariant to orthogonal rotation, so this does not trivialize the detector, but it only genuinely reduces dimensionality at $M = 5$ in the hyperparameter grid of Section 5). These four detectors are constructed from structurally different statistics, isolation depth, local density, Mahalanobis distance, and diffusion reconstruction difficulty, and we do not assume any one of them is a monotone function of the embedding's Euclidean norm; Section 4.4's mechanism argument only requires that each detector tend, on average, to assign higher atypicality to larger perturbations somewhere in the embedding, which is a weaker and more defensible premise than strict norm-monotonicity, and which the shared empirical pattern across all four detectors in Tables 1–3 is itself evidence for. All four are fit causally on an expanding window (minimum 250 observations) and retrained every 20 folds on history up to the current origin only; none has access to any data at or after the origin being scored.

3.4. Evaluation

We use nested nonoverlapping walk-forward folds with step size H over an expanding window starting at 250 observations. The accuracy-gap statistic is defined by a median split on the atypicality score, where the median at fold i is computed causally from the atypicality scores of folds strictly before i, never from the full sample; its significance is assessed with a 2000-resample permutation test, one-sided against the alternative that the gap is positive (high-atypicality accuracy exceeds low-atypicality accuracy), so a reported p close to 1, as for several cells of Table 3, indicates a point estimate in the opposite direction rather than a computational error. We combine p-values across the seven-asset panel with Fisher's method, noting that this assumes independence across assets, which does not hold exactly for financial return series; under positive dependence, Fisher's method is anti-conservative (it understates the combined p-value), so the panel-level p-values we report below should be read as more optimistic than a dependence-robust combination

would give. Separately, we hold out the last 20% of each asset's history, untouched during all model development, and evaluate it exactly once.

4. The Error: Discovery, Diagnosis, and Fix

4.1. The original result

Using the method of Section 3, we found a substantial, statistically significant gap in forecast accuracy between high- and low-atypicality origins, consistently across all four detectors and all seven assets (Table 1). The diffusion-ELBO detector, for example, showed a +8.3 percentage-point accuracy gap on Brent crude ($p < 0.0001$), and the three classical detectors showed comparable or larger gaps on the same and other assets. A genuine single-shot holdout test, touched for the first time only after model development was frozen, reproduced the effect at the panel level, with a Fisher-combined $p = 3.8 \times 10^{-5}$ across the seven assets. We additionally proposed and validated a dynamical-systems explanation involving local Lyapunov exponents, which itself replicated on a second, independent holdout and proved robust to parameter choices and to splitting the sample by calendar era. Section 4.5 returns to what happened to this explanation once the indexing error was corrected.

Table 1: Accuracy gap by detector and asset, uncorrected pipeline (percentage points; p-values from a 2000-resample permutation test).

Asset	Diffusion	Isolation Forest	LOF	Mahalanobis-PCA
Brent	8.3, $p < 0.0001$	13.5, $p < 0.0001$	13.4, $p < 0.0001$	13.1, $p < 0.0001$
S&P 500	2.9, $p = 0.113$	4.8, $p = 0.028$	5.3, $p = 0.013$	5.1, $p = 0.016$
Gold	4.9, $p = 0.022$	11.4, $p < 0.0001$	7.4, $p = 0.002$	8.8, $p < 0.0001$
Bitcoin	11.0, $p < 0.0001$	11.9, $p < 0.0001$	12.0, $p < 0.0001$	12.6, $p < 0.0001$
Ethereum	6.5, $p = 0.027$	14.4, $p < 0.0001$	8.7, $p = 0.003$	13.8, $p < 0.0001$
Silver	10.9, $p < 0.0001$	10.9, $p < 0.0001$	11.1, $p < 0.0001$	13.8, $p < 0.0001$
WTI Crude	10.1, $p < 0.0001$	7.5, $p = 0.002$	10.2, $p < 0.0001$	9.3, $p < 0.0001$

4.2. The surrogate-data control that exposed the problem

The problem surfaced during what we intended as a routine robustness check of the dynamical-systems explanation, not of the core claim. We generated AAFT surrogate return series, matching the real data's amplitude

distribution exactly and its linear power spectrum approximately while randomizing phases to destroy any nonlinear temporal structure, and re-ran the local-Lyapunov analysis on them. The purely linear surrogates reproduced, and in fact exceeded, the real data's effect size, for both a cheap k-NN atypicality proxy and the production diffusion-ELBO detector. Taken on its own, that result does not require positing a coding error; the surrogate-data literature is full of cases where apparent nonlinear structure turns out to be linear. It did, however, prompt us to apply the same control to the paper's actual central claim, the atypicality-conditioned accuracy gap, rather than only to the auxiliary Lyapunov statistic.

4.3. The decisive control: pure white noise

We generated literal i.i.d. Gaussian white noise, a process with, by construction, no genuine forecastable structure of any kind, and ran the identical pipeline on it. The pipeline reported a significant atypicality-conditioned accuracy gap of 5–7 percentage points, with permutation p-values as low as the resolution floor of a 2000-resample test, on every one of three initial noise draws; this was extended to ten trials across two assets in Section 5, with the same result each time. A process with zero genuine predictive structure cannot produce a genuine accuracy edge of that size by definition, so the only remaining explanation is a defect in the measurement procedure itself.

4.4. Root cause

In plain terms, the pipeline's "most recent known return," the feature used to judge how unusual today looks and which historical days resemble it, was not actually known yet at the moment the forecast was made. It was one day into the future the model was supposedly predicting.

Table 2 fixes the notation used from here on: j indexes an origin in the pipeline's internal, zero-based array (the quantity in the code itself), while p, also an array index, marks the position the pipeline treats as "today's price." Calendar time t used elsewhere in the paper (Section 3.2's x_t) corresponds to array index $j + (M-1)\tau$, the position the embedding's own most recent return ends at; under the corrected convention this is $t = p - \tau$, one step before "today's price" $p = j + M\tau$, which is exactly the separation the bug collapsed. We use j and p rather than t in this section specifically because the bug is a one-step discrepancy between two candidate array positions, and conflating them with a single calendar-time variable is exactly the ambiguity that let the bug go undetected.

Table 2: Notation used in Section 4.4.

Symbol	Meaning
P_t	Price at calendar/array position t
r_t, `log_ret`$[t]$	Return ending at position t, $r_t = \log(P_t/P_{t-1})$
j	Origin index in the pipeline's internal array (zero-based)
M, τ	Embedding length and delay ($M = 3$, $\tau = 1$)
p	Array index the pipeline treats as "today's price"
$x_j^{(0)}$	Embedding's most recent (zero-lag) component for origin j
H	Forecast horizon ($H = 3$)
y_j	Forecast target for origin j: the H-step return beginning at p

With origin index j, embedding length M, and delay τ, the project's convention located "today's price" at array index $p = j+(M-1)\tau$ and defined the forecast target as the H-step return beginning at p. Throughout this section, $x_j^{(0)}$ denotes the embedding's most recent component *before* the scale division of Section 3.2: that division is a positive constant, shared by every component of every origin's embedding, applied for magnitude normalization across detectors, and it does not change which array index is read into any component, only its units, so the indexing argument below is unaffected by whether it is stated for the raw or the scaled component. In raw form, $x_j^{(0)}$ is constructed from `log_ret`$[j + (M - 1)\tau]$, which is exactly `log_ret`$[p]$: the first of the H returns summed to produce the forecast target itself (the target itself is never scaled by σ, since it is a realized price return, not a model input). The scaled embedding component actually used by the detectors is $x_j^{(0)}/\sigma$, proportional to `log_ret`$[p]$ by the same fixed positive constant for every origin, so it carries the identical sign and the identical index-overlap problem. The correct convention is $p = j + M\tau$. Under this convention, the embedding's most recent return is `log_ret`$[p - \tau]$, the return that ends exactly at today's price, `prices`$[p]$; the forecast target is the return that begins at `prices`$[p]$ and extends forward. The two are adjacent in the return index, so no day of history is skipped, but the target return itself is entirely unrealized as of today, which is what the original, off-by-one convention failed to guarantee. Figure 1 illustrates both conventions on a common return-index axis.

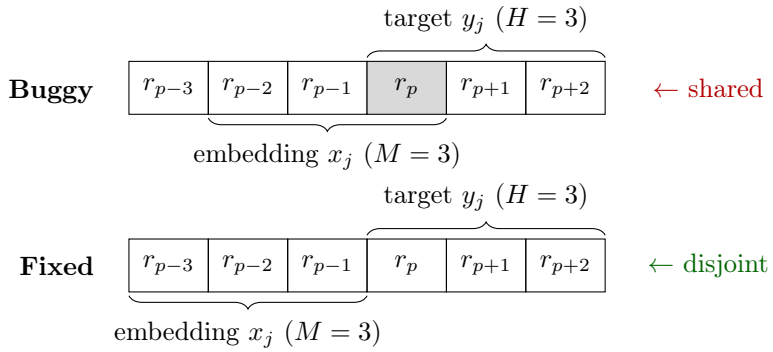


Figure 1: The embedding (bottom brace) and the forecast target (top brace) on a shared return-index axis. Under the buggy convention (top), the embedding's most recent component, r_p (shaded), is also the first return summed into the target: the model is given, as input, one of the three numbers it is being asked to predict. Under the fixed convention (bottom), the embedding ends at r_{p-1} and the target begins at r_p; the two sets of returns no longer overlap.

This also explains why the leak inflated accuracy specifically on days flagged as atypical, rather than uniformly across all origins. The leaked return, `log_ret`$[p]$, enters both the embedding norm, dominating it whenever $|$`log_ret`$[p]|$ is itself large relative to the embedding's other two components (Section 3.3 states the atypicality-score relationship more carefully: we do not assume any detector is a strict monotone function of the norm, only that each tends to score large perturbations as more atypical), and the first of the three summands making up the $H = 3$-day target. When $|$`log_ret`$[p]|$ is itself large, an origin is simultaneously pushed toward the high-atypicality tail of the detector's score and, because a large first summand increasingly dominates the three-day sum it belongs to, pushed toward a target whose sign agrees with the leaked return's own sign. On origins where $|$`log_ret`$[p]|$ is small, by contrast, the detector has little reason to flag the origin as atypical, and the sign of the three-day target is instead decided by the two unleaked future returns the model has no access to, leaving accuracy near chance. The leak and the atypicality label were, in other words, driven by the same quantity, which is why conditioning on atypicality looked like it was isolating a genuinely more forecastable regime rather than simply isolating the origins where the leak itself was largest.

This can be made precise under a simple idealization. Model returns as i.i.d. Gaussian, $r_t \sim \mathcal{N}(0, \sigma^2)$, and write the leaked target as $y_p = r_p + R_+$,

where r_p is the leaked component and $R_+ = \sum_{h=1}^{H-1} r_{p+h} \sim \mathcal{N}(0, (H-1)\sigma^2)$ is independent of r_p. Then $\mathrm{Cov}(r_p, y_p) = \mathrm{Var}(r_p) = \sigma^2$ (the R_+ term contributes nothing to the covariance, by independence), while $\mathrm{Var}(y_p) = \mathrm{Var}(r_p) + \mathrm{Var}(R_+) = H\sigma^2$. Since (r_p, y_p) are jointly Gaussian with correlation $\rho = \mathrm{Cov}(r_p, y_p)/\sqrt{\mathrm{Var}(r_p)\mathrm{Var}(y_p)} = \sigma^2/(\sigma \cdot \sigma\sqrt{H}) = 1/\sqrt{H}$, a standard identity for bivariate Gaussians gives

$$\mathbb{P}\big(\mathrm{sign}(r_p) = \mathrm{sign}(y_p)\big) = \frac{1}{2} + \frac{1}{\pi}\arcsin\left(\frac{1}{\sqrt{H}}\right). \tag{1}$$

For $H = 3$, this evaluates to $\approx 69.6\%$, close to the 68.3% unconditional sign-agreement rate measured directly on real Brent data in check 3 of Section 5; the small gap is consistent with real returns not being exactly i.i.d. Gaussian. This is the unconditional rate, over all origins. Conditional on the leaked return's magnitude, $|r_p| = c$, the same Gaussian calculation gives $\mathbb{P}(\mathrm{sign}(r_p) = \mathrm{sign}(y_p) \mid |r_p| = c) = \Phi(c/(\sigma\sqrt{H-1}))$, which is monotonically increasing in c and tends to 1 as c grows; since all four detectors assign higher atypicality scores to embeddings with larger Euclidean norm, and $|r_p|$ is the dominant term in that norm, conditioning on high atypicality is, under this idealization, equivalent to conditioning on large c, which pushes the sign-agreement probability above the unconditional baseline computed above. This is the formal counterpart of the qualitative mechanism described above, not a claim that real returns are Gaussian; it is offered to show that the regime-concentration effect follows from the leak's definition alone, without appeal to any genuine market structure.

One link remains to close: the pipeline's forecast is not $\mathrm{sign}(r_p)$ directly, but the sign of the mean H-day return realized by the $K = 7$ historical analogs nearest to the current embedding. The same index convention, however, applies identically to every historical origin, not only the current one, so each candidate analog's own embedding carries its own leaked return, and each analog's own realized target begins with that same leaked return. An analog selected because its embedding is close to the current one is, for that reason, also likely to have a leaked return of similar sign and comparable magnitude to r_p; the analogs' averaged realized target therefore tends to carry the same sign as r_p as well, for the same reason the current origin's own target does. This holds whether the nearest analogs are selected by raw embedding distance or, as in our pipeline, by distance in the diffusion model's hidden-layer feature space (Section 3.3), since that representation is trained to reconstruct the embedding and so preserves its leaked component's sign to

a close approximation. The leak therefore reaches the final forecast through the analog average, not only through the current origin's own embedding.

This qualitative argument also has a quantitative counterpart under the same idealization as Equation (1), which we record because it closes a gap the single-origin result alone leaves open: that equation gives the probability the current origin's own target agrees in sign with its leaked component, not the probability the K-analog ensemble forecast agrees with the realized target. If every selected analog shares the same leaked component r_p exactly (the idealization above) with independent unleaked continuations $R_{+,k} \sim \mathcal{N}(0, (H-1)\sigma^2)$, the analog average is $\bar{y}_K = r_p + \frac{1}{K}\sum_{k=1}^{K} R_{+,k}$, whose unleaked term has variance reduced by a factor of K; the same bivariate-Gaussian argument then gives, conditional on $|r_p| = c$, $\mathbb{P}(\text{sign}(\bar{y}_K) = \text{sign}(r_p) \mid |r_p| = c) = \Phi(\sqrt{K}\, c/(\sigma\sqrt{H-1})) \equiv q_K(c)$, and since $\bar{y}_K$ and the current origin's own target y_p are conditionally independent given r_p, the ensemble's forecast accuracy is $q(c)\, q_K(c) + (1 - q(c))(1 - q_K(c))$, where $q(c) = \Phi(c/(\sigma\sqrt{H-1}))$ is the single-origin probability from Equation (1). We verified this expression against direct Monte Carlo simulation of the idealization (agreement to three significant figures), and it behaves as the qualitative argument above suggests: it equals 0.5 at $c = 0$ and, because q_K grows faster than q in c for $K > 1$, approaches 1 faster than the single-origin probability alone as c grows, so averaging over $K = 7$ analogs sharpens, rather than dilutes, the leak's effect on the final forecast.

4.5. The fate of the Lyapunov explanation

We now ask whether the dynamical-systems explanation (Section 4.1) is a second casualty of the same indexing error, or a separate problem. It is the latter. Local Lyapunov exponents were estimated, for each origin j with causal nearest neighbor k (excluding the Theiler window) and a forward horizon Δt equal to the forecast horizon H, as the log-rate of divergence of the two trajectories, $\lambda_j = \frac{1}{\Delta t}\log\big(\|x_{j+\Delta t} - x_{k+\Delta t}\|/\|x_j - x_k\|\big)$, with more negative values indicating faster local contraction (greater short-run predictability under the dynamical-systems reading) and values near zero or positive indicating local divergence. The local-Lyapunov-exponent calculation uses the origin index j and a forward offset directly; it never reads the mis-specified index p at all, so it was never affected by the bug in the first place. Rerunning it against the corrected pipeline's atypicality scores confirms this directly: the atypical-versus-typical gap in local Lyapunov exponents is essentially unchanged from the original, uncorrected run on five of seven assets

(for example, Brent: -0.166 before correction, -0.166 after), and remains of the same sign and order of magnitude on the remaining two, Bitcoin (-0.206 before, -0.216 after) and Ethereum (-0.203 before, -0.179 after), a gap we attribute to the usual combination of updated price history between runs and shifted fold boundaries near the end of these two shorter series, not to the indexing fix. All seven remain significant at $p < 0.001$ in both runs. The index fix, in other words, does not explain away the Lyapunov result.

That does not rehabilitate it. The surrogate-data control in Section 4.2 already showed, by a different route, that this same gap is reproduced, and in fact exceeded, by AAFT surrogates with no genuine dynamics of any kind. Taken together, the two results describe two independent problems rather than one shared cause: the accuracy-gap claim was an artifact of the index error and disappears once it is fixed; the Lyapunov explanation was never touched by that error, but is a separate pattern exposed as an artifact by the surrogate-data test rather than by the index fix: because a purely linear AAFT surrogate, with no genuine nonlinear dynamics by construction, reproduces and exceeds the same local-Lyapunov gap, the result cannot be attributed to genuine deterministic chaos, whatever its precise cause turns out to be. We consider a finite-sample geometric bias in nearest-neighbor-based divergence-rate estimation the most likely specific explanation, consistent with known limitations of this estimator in the nonlinear time-series literature [27, 16], but the surrogate test alone does not distinguish that explanation from other linear-structure accounts, and we do not claim to have isolated the precise mechanism beyond ruling out genuine nonlinear dynamics. We flag this distinction explicitly because conflating the two, concluding that fixing the index error also explains the Lyapunov result, or that the Lyapunov result's survival means the index fix was somehow incomplete, would both be incorrect.

4.6. Why none of the standard safeguards caught it

The error sits in every fold's feature-construction code identically, in both the training and the held-out portions of the data, because it is a fixed, deterministic property of the index arithmetic rather than something learned from or overfit to any particular sample. That is precisely why each of the three safeguards this literature relies on most heavily missed it. Walk-forward cross-validation protects against overfitting to one historical window, but offers no protection against a structural bias present identically in every window. A genuine single-shot holdout protects against re-using information

learned during model selection, but offers no protection against a bias baked into the feature-construction code itself, which is applied, by construction, the same way to the holdout period as to everything else. Permutation testing evaluates the null hypothesis that forecast accuracy is statistically independent of the atypicality label; shuffling the labels correctly destroys that coupling and produces a null distribution of gaps centered near zero in every one of the shuffled resamples. The test nonetheless offers no protection here because the null hypothesis it tests is, in this case, genuinely false: the leak ties the atypicality score and the target's leaked component to the same underlying quantity, so accuracy really is statistically dependent on the atypicality label, identically in every fold. The small p-value the test returns is a correct answer to the question it asks, dependence is real, not a false positive; the error is in treating statistical dependence, by itself, as evidence of genuine forecastable structure rather than of a confound.

Each of these three tools is, individually, considered strong evidence in the forecasting literature, and none of them is designed to catch a fixed arithmetic bias in feature construction, because none of them varies the feature-construction code itself. A fourth tool does: comparing the pipeline's output on real data against its output on data engineered to contain no real structure at all (Section 4.3, formalized as a general protocol in Section 7.2) tests the feature-construction code directly, rather than the sample it happens to be run on.

5. Verification

Given what the diagnosis implies, the central claim of a multi-month investigation turning out to be an artifact, we did not accept it on the strength of any single check. The six checks below span increasing levels of pipeline complexity, from a closed-form algebraic identity to the full production pipeline. Checks 1 and 3 are two representations of the same underlying fact rather than independent evidence, and are listed separately only because they operate at different levels of abstraction. Full detail and numbers are available in the accompanying code repository (see the Data and Code Availability statement).

1. **Exact algebraic identity.** We hand-derived and numerically confirmed that $x_j^{(0)}$ (the embedding's raw, pre-scaling component defined above) equals `log_ret`$[p]$ exactly, not approximately, on a tiny synthetic price array.

2. **Generalization across the real series.** Using the project's own code, we repeated the algebraic check over 33 origins spread across real Brent history: the overlap held universally under the old formula and never under the corrected one.
3. **A pipeline-free, model-free direct test.** The sign of $x_j^{(0)}$ matched the sign of the actual future return in 68.3% of origins under the old convention, against 50.1%, consistent with chance, under the fix. That 68.3% rather than roughly 100% is itself consistent with the diagnosis, since the leaked component is only one of three summands making up the target.
4. **A null control repeated across ten independent draws** of synthetic i.i.d. Gaussian white noise, matched to Brent's and S&P 500's sample length and return volatility (three draws at Brent's parameters, seven more at S&P 500's, with different random seeds; no real price data is used in this check). Under the fixed pipeline, the mean permutation p-value across all ten draws was 0.496, consistent with the theoretical expectation of 0.5 under a true null; one of the ten crossed the nominal 5% significance threshold ($p = 0.037$), which is itself unremarkable at that sample size.
5. **A fully independent, from-scratch script** with no imports from the project's own codebase, using a fresh data download and hand-written embedding and index arithmetic, repeated the same algebraic check over 47 origins: again, universal overlap under the old formula and none under the fix. Agreement between this standalone script and check 2 rules out a shared misreading of the code baked into the investigation's own tooling.
6. **A power test with a known, designed-in signal.** We built a synthetic series with an explicit mean-reversion rule: after any return exceeding one standard deviation, the next three returns are deliberately biased in the opposite direction. The fixed pipeline correctly detected this designed-in signal: low-atypicality-subgroup accuracy was 52.2%, versus 58.2% for the high-atypicality subgroup, within this single synthetic series (gap $p = 0.011$, diffusion-ELBO detector), confirming that the fix did not simply desensitize the method.

We also ran a broader leakage audit covering data-feature temporal ordering, embedding normalization, train/test shuffling, and detector self-inclusion. Two of these checks were direct re-verifications that the indexing

error above was gone and stayed gone (target-feature alignment and residual embedding overlap, already covered by checks 1, 3, and 5), rather than searches for a separate issue, so we do not count them as new findings. The audit's one genuinely new result is that the embedding's scale normalization is computed over the full series rather than causally. We disclose it here and report the complete check, extended from an initial two-asset spot check to the full panel: all seven assets, all four detectors, both the full-sample scale actually used in the paper and a corrected causal (growing-window) variant, 56 gap estimates in total. Every diffusion-detector cell is non-significant under both conventions (Brent: +2.6pp moves to −0.4pp; S&P 500: −1.2pp to −2.9pp; Gold: −5.4pp to −6.4pp; Bitcoin: −1.7pp to +0.1pp; Ethereum: +1.2pp to −0.6pp; Silver: +1.4pp to +0.9pp; WTI Crude: −1.0pp to +0.6pp, matching Table 3's full-sample-scale column), and across all 56 cells exactly the same single cell already flagged in Table 3, Ethereum/Mahalanobis-PCA, is significant at $p < 0.05$ under both the full-sample scale ($p = 0.025$) and the causal scale ($p = 0.017$); every other cell is non-significant under both conventions. We read this as the same marginal, likely-chance cell reappearing under a second normalization choice, not as a second, independent leakage channel. Separately, as direct empirical evidence for the mechanism argued analytically in Section 4, we computed the Spearman correlation between each detector's atypicality score, on the corrected pipeline's embedding, and the magnitude of that embedding's own most recent raw-return component, $|\texttt{log_ret}[j + (M-1)\tau]|$, i.e. the exact quantity the buggy convention reads directly into the forecast target, across all 28 asset-detector combinations (7 assets, 4 detectors, full-sample scale). An earlier version of this check correlated detector scores against the magnitude of the return one step later instead (the fixed convention's target, not the embedding's own component), which mostly measures ordinary volatility clustering between adjacent days rather than the detector's sensitivity to its own input; correcting this, every one of the 28 correlations is positive, substantially stronger than the misspecified version (Spearman ρ ranging from 0.41 to 0.61, median 0.53, versus 0.09 to 0.32 in the earlier, incorrect check), and significant beyond $p < 10^{-38}$ in every case. This is direct confirmation that all four structurally different detectors, not only ones provably monotonic in the embedding norm, score larger values of the exact leaked component as more atypical.

To check whether the leak-amplification mechanism is specific to the paper's own hyperparameter choices ($M = 3$, $\tau = 1$, $H = 3$) or holds more generally, we ran a controlled grid over 18 hyperparameter combinations

($M \in \{2, 3, 5\}$, $\tau \in \{1, 2\}$, $H \in \{1, 3, 5\}$), each evaluated at three leak severities, `leak_steps` $\in \{0, 1, 2\}$ (0 reproduces the fixed convention; 1 reproduces the paper's actual bug; 2 is a more severe variant), applying the buggy index convention uniformly to every origin's own target and to every one of its analogs' targets (not only the origin being scored, which is what actually makes the leak propagate through the K-neighbor average), on synthetic Gaussian AR(1) data ($\phi = 0.15$, matched to Brent's volatility) with three classical detectors (diffusion retraining at every grid point was not computationally feasible for a grid this size). For each leak severity, averaging its mean gap over all 18 hyperparameter combinations and all three detectors (54 cell-detector pairs, 5 seeds each): a leak-free origin (`leak_steps` $= 0$) gives a mean gap of +0.2pp, a one-step leak matching the paper's actual bug gives +7.1pp, and a two-step leak gives +7.7pp, replicated across every one of the 18 (M, τ, H) combinations tested, not only the paper's own choice of hyperparameters.

Finally, we extended the paper's Gaussian-IID white-noise null control (Section 4.3) to three additional processes real returns resemble more closely, Student-t ($\nu = 4$) IID, Gaussian AR(1) ($\phi = 0.15$), and GARCH(1,1) ($\alpha = 0.08$, $\beta = 0.90$), each run through the identical diffusion pipeline under both the buggy and fixed index convention, 5 independent seeds per process (mean gap $\pm$ one seed-to-seed standard deviation: Student-t buggy $+6.3 \pm 2.6$pp, fixed -2.0 ± 3.6pp; AR(1) buggy $+4.6 \pm 3.8$pp, fixed $+3.0 \pm 3.1$pp; GARCH(1,1) buggy $+3.5 \pm 1.9$pp, fixed $+1.7 \pm 2.4$pp). We distinguish two of these three from a true zero-predictability null before interpreting the comparison: Student-t IID and GARCH(1,1) with a zero-mean conditional distribution both have $\mathbb{E}[\text{sign}(y_p) \mid \mathcal{F}_{p-1}] = 0$ exactly, so any gap under the fixed convention is attributable only to sampling noise. Gaussian AR(1) with $\phi = 0.15$ does not have this property: under the fixed convention, the causally known return r_{p-1} carries genuine linear information about the target, $\mathbb{E}[y_p \mid r_{p-1}] = (\phi + \phi^2 + \phi^3)\, r_{p-1} \approx 0.176\, r_{p-1}$, and by the same bivariate-Gaussian conditioning as Equation (1), large $|r_{p-1}|$ both inflates the embedding norm and strengthens this genuine directional signal. A positive fixed-convention gap on AR(1) is therefore not unexpected or a sign of residual leakage; it is what a correctly functioning pipeline should show on a series with real, if weak, linear autocorrelation. Judged against that distinction, the strict-null processes behave as the mechanism predicts: the buggy convention's mean exceeds the fixed convention's by a wide margin on Student-t, with the fixed mean close to zero as expected, while GARCH(1,1)

shows the same direction more weakly (gap between means smaller relative to seed-to-seed spread) and would benefit from more than 5 seeds before drawing a firm conclusion. We report this honestly as a noisier, lower-powered replication than the original Gaussian-IID control, not a second clean one; the Monte Carlo grid above, run on a much larger number of configurations, is the stronger piece of generalization evidence.

6. Results After Correction

Table 3: Accuracy gap by detector and asset, corrected pipeline (percentage points; p-values from a 2000-resample permutation test).

Asset	Diffusion	Isolation Forest	LOF	Mahalanobis-PCA
Brent	$+2.6$, $p = 0.165$	$+1.8$, $p = 0.252$	$+0.2$, $p = 0.477$	$+0.7$, $p = 0.398$
S&P 500	-1.2, $p = 0.692$	-0.6, $p = 0.605$	-1.6, $p = 0.747$	-1.1, $p = 0.677$
Gold	-5.4, $p = 0.985$	$+0.9$, $p = 0.346$	-0.5, $p = 0.583$	-1.0, $p = 0.640$
Bitcoin	-1.7, $p = 0.720$	$+0.6$, $p = 0.422$	-2.9, $p = 0.823$	-1.4, $p = 0.689$
Ethereum	$+1.2$, $p = 0.352$	$+3.6$, $p = 0.135$	$+1.1$, $p = 0.372$	$+6.2$, $p = 0.025$
Silver	$+1.4$, $p = 0.302$	-2.0, $p = 0.802$	-1.8, $p = 0.774$	-1.8, $p = 0.762$
WTI Crude	-1.0, $p = 0.654$	-0.3, $p = 0.551$	-0.2, $p = 0.518$	-0.1, $p = 0.516$

No cell in Table 3 that was significant in Table 1 remains so after the fix, and several point estimates change sign. Across all 28 detector-asset cells, exactly 1 falls below $p = 0.05$ (Ethereum/Mahalanobis-PCA, $p = 0.025$). Because the four detectors share the same underlying embedding and the seven assets are not independent of one another, these 28 tests are correlated rather than independent, so we do not report a formal expected false-positive count; qualitatively, one marginal cell out of 28 against a nominal 5% threshold is consistent with chance and does not, on its own, constitute evidence of a surviving effect.

The genuine single-shot holdout test, re-run with the corrected pipeline, tells the same story at the panel level. Every asset is individually non-significant, and Fisher's combined p-value across the seven-asset panel is **0.288**, against the original, uncorrected pipeline's $p = 3.8 \times 10^{-5}$ on the identical protocol. The strongest-looking evidence in the original investigation collapses, once the leak is removed, to a result indistinguishable from chance.

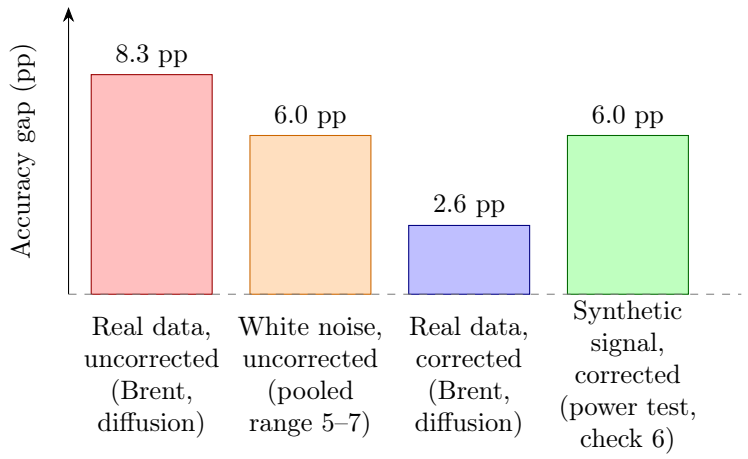


Figure 2: Accuracy gap across four regimes, each number taken directly from a result already reported in the text: the uncorrected pipeline shows a large gap on real data that a pure-noise null control (no genuine structure by construction) already reproduces at comparable size; the corrected pipeline's gap on real data collapses toward zero, while its gap on a synthetic series with a designed-in signal remains intact, confirming the fix removed the artifact without destroying the method's ability to detect genuine structure.

7. Discussion

7.1. Why this is worth publishing

We think a null result reached through careful, multiply-verified diagnosis of a subtle and previously undocumented methodological error is more useful to the forecasting community than the original, incorrect, positive finding would have been. The error is structural to a widely used class of method, delay embedding paired with auxiliary novelty or anomaly scoring, and is not specific to our particular implementation choices, so it is likely to recur wherever the same convention is used. We also show, concretely, that the standard toolkit for guarding against overfitting in this literature, walk-forward validation, permutation testing, and even genuine out-of-sample holdout testing, did not detect it. That is itself an actionable finding about the limits of those safeguards for this specific failure mode, and it echoes a broader point made by Bailey et al. [2] about the gap between formal rigor and actual protection against overfitting in backtest-driven research. Finally, we offer a lightweight, generally applicable diagnostic, a null-noise control paired with surrogate-data testing, that would have caught this class of error cheaply, before a researcher spends months building an elaborate positive-finding narrative on

top of it, as we did.

7.2. A recommended diagnostic protocol

We recommend that any study pairing delay embedding with an auxiliary detection score for financial, or other, time-series forecasting run two checks before reporting a positive result. The first is a null-data control: apply the identical pipeline to synthetic data with provably zero real structure (i.i.d. noise matched to the real series' scale is sufficient), and confirm that the reported effect does not appear. The second is a surrogate-data control: apply the identical pipeline to AAFT, or comparable, surrogates of the real series, to separate genuine structure from an artifact that matched linear noise can reproduce on its own. Both checks are inexpensive relative to the modeling effort typically invested in this class of study, and together they would have caught the error documented here within hours rather than months.

7.3. Limitations

This paper documents one specific error in one specific implementation of a standard method family. We are not claiming the error is present in any other published study; we are claiming that the general class of convention it belongs to, using the embedding's most-recent-return component without an explicit, verified alignment check against the forecast target's own starting index, carries this risk, and that the diagnostic protocol we propose is a cheap, general-purpose safeguard whether or not any specific paper happens to contain this exact error.

8. Conclusion

We set out to validate the claim that market states flagged as locally atypical by an unsupervised novelty detector are more forecastable than typical states. The claim passed every standard validation step used in this literature. It did not survive a null-data control, because the apparent effect was, in fact, a one-step index error that gave the forecasting model partial access to its own target. Section 5's six checks confirm this diagnosis, and show that the corrected pipeline retains the ability to detect genuine designed-in signal, which rules out the possibility that the correction simply broke the method. We offer the diagnostic protocol that exposed the error, a null-noise control paired with a surrogate-data test, both standard tools in adjacent literatures

but, to our knowledge, not yet standard practice in delay-embedding-based financial forecasting, as this paper's main contribution.

Data and Code Availability

All code needed to reproduce every result in this paper, including the original (uncorrected) pipeline, the fix, and all six verification checks in Section 5, together with a documented walkthrough of each result and step-by-step reproduction instructions, will be made available to reviewers and readers on request to the corresponding author during the review process, and deposited in a permanent, versioned public archive (for example via Zenodo, with a DOI) upon acceptance. All price data is downloaded at run time from Yahoo Finance and is not redistributed.

CRediT Authorship Contribution Statement

Ramin Buzarpur: Conceptualization, Software, Methodology, Formal analysis, Investigation, Data curation, Writing – original draft.
Armin Buzarpur: Methodology, Formal analysis (dynamical-systems and control-theoretic framing), Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of Generative AI Use

During the preparation of this work, the authors used an AI assistant (Claude, Anthropic) to proofread drafts and improve language and clarity. The authors conceived the study, performed the analysis, wrote the original text, and reviewed and edited all content before submission, and take full responsibility for the content of the publication.

References

[1] Alessandrini, S., Delle Monache, L., Sperati, S., & Nissen, J. (2015). A novel application of an analog ensemble for short-term wind power forecasting. *Renewable Energy*, 76, 768–781.

[2] Bailey, D. H., Borwein, J. M., López de Prado, M., & Zhu, Q. J. (2014). Pseudo-mathematics and financial charlatanism: The effects of back-test overfitting on out-of-sample performance. *Notices of the American Mathematical Society*, 61(5), 458–471.

[3] Barnett, W. A., & Serletis, A. (2000). Martingales, nonlinearity, and chaos. *Journal of Economic Dynamics and Control*, 24(5–7), 703–724.

[4] Bergmeir, C., Hyndman, R. J., & Koo, B. (2018). A note on the validity of cross-validation for evaluating autoregressive time series prediction. *Computational Statistics & Data Analysis*, 120, 70–83.

[5] Breunig, M. M., Kriegel, H.-P., Ng, R. T., & Sander, J. (2000). LOF: Identifying density-based local outliers. In *Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data* (pp. 93–104).

[6] Coqueret, G. (2024). Forking paths in financial economics. *arXiv preprint* arXiv:2401.08606.

[7] Delle Monache, L., Nipen, T., Liu, Y., Roux, G., & Stull, R. (2011). Kalman filter and analog schemes to post-process numerical weather predictions. *Monthly Weather Review*, 139, 3554–3570.

[8] Delle Monache, L., Eckel, F. A., Rife, D. L., Nagarajan, B., & Searight, K. (2013). Probabilistic weather prediction with an analog ensemble. *Monthly Weather Review*, 141, 3498–3516.

[9] Gelman, A., & Loken, E. (2014). The statistical crisis in science. *American Scientist*, 102(6), 460–465.

[10] Good, P. I. (2005). *Permutation, Parametric, and Bootstrap Tests of Hypotheses* (3rd ed.). Springer Series in Statistics. Springer-Verlag.

[11] Hamill, T. M., & Whitaker, J. S. (2006). Probabilistic quantitative precipitation forecasts based on reforecast analogs: Theory and application. *Monthly Weather Review*, 134(11), 3209–3229.

[12] Harvey, C. R., Liu, Y., & Zhu, H. (2016). ...and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.

[13] Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33, 6840–6851.

[14] Hou, K., Xue, C., & Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5), 2019–2133.

[15] Inglada-Pérez, L. (2020). A comprehensive framework for uncovering non-linearity and chaos in financial markets: Empirical evidence for four major stock market indices. *Entropy*, 22(12), 1435.

[16] Kantz, H., & Schreiber, T. (2004). *Nonlinear Time Series Analysis* (2nd ed.). Cambridge University Press.

[17] Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9), 100804.

[18] Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4), Article 15.

[19] Ledoit, O., & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.

[20] Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation forest. In *2008 Eighth IEEE International Conference on Data Mining* (pp. 413–422).

[21] Nikolopoulos, S. D. (2026). Spurious predictability in financial machine learning. *arXiv preprint* arXiv:2604.15531.

[22] Petropoulos, F., Apiletti, D., Assimakopoulos, V., Babai, M. Z., Barrow, D. K., Ben Taieb, S., et al. (2022). Forecasting: Theory and practice. *International Journal of Forecasting*, 38(3), 705–871.

[23] Schreiber, T., & Schmitz, A. (1996). Improved surrogate data for nonlinearity tests. *Physical Review Letters*, 77, 635–638.

[24] Schreiber, T., & Schmitz, A. (2000). Surrogate time series. *Physica D*, 142, 346–382.

[25] Takens, F. (1981). Detecting strange attractors in turbulence. In *Dynamical Systems and Turbulence, Warwick 1980* (pp. 366–381). Springer.

[26] Theiler, J. (1986). Spurious dimension from correlation algorithms applied to limited time-series data. *Physical Review A*, 34(3), 2427.

[27] Theiler, J., Galdrikian, B., Longtin, A., Eubank, S., & Farmer, J. D. (1992). Testing for nonlinearity in time series: The method of surrogate data. *Physica D*, 58, 77–94.

[28] Vogl, M., & Roetzel, P. G. (2022). Chaoticity versus stochasticity in financial markets: Are daily S&P 500 return dynamics chaotic? *Communications in Nonlinear Science and Numerical Simulation*, 108, 106218.

[29] White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097–1126.